By TOUCH DEN · November 11, 2026

Access Control Basics



Video 1 · Authorization, Least Privilege, and Separation of Duties

Security Fundamentals · Section 7

In this video, you will learn:

- Authentication vs. authorization
- The principle of least privilege
- Separation of duties

Authentication vs. authorization

- ***Authentication:*** *"who are you?" Happens once, at the door. (Lesson 6.)*
- ***Authorization:*** *"what are you allowed to do?" Happens on* ***every action****, after you're known.*

At an airport: your ***passport*** *proves who you are (authentication); your* ***boarding pass*** *says which flight and seat (authorization).*

A huge share of breaches are ***broken authorization****: a logged-in user reaches data they should never have, often by changing a number in a web address.*

Least privilege

Give each person, program, and device only the access it needs, and no more.

The single most important idea in access control.

- ***Limits damage:*** *a phished account gains only its small slice, not everything. Turns a disaster into an incident.*
- ***Limits mistakes:*** *you can't delete what you can't touch.*
- ***Shrinks the attack surface.***

Even admins browse and read email as an ***ordinary user****, elevating only for the task that needs it.*

Separation of duties

No single person can complete a sensitive process alone.

- *Classic example: the person who **requests** a payment is not the one who **approves** it.*
- *To commit fraud you'd now have to corrupt two people, much harder and more likely to be noticed.*

Pairs with least privilege: least privilege limits one role's power; separation of duties splits the most dangerous powers across roles.

Recap: the core principles

- **Authentication** = who you are; **authorization** = what you may do, checked on every action.
- **Least privilege:** only the access needed; limits damage and mistakes.
- **Separation of duties:** split sensitive actions so no one person acts alone.

Next video: the models, roles, and why access rots over time.

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Video 2 · Models, Roles, and Privilege Creep

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In this video, you will learn:

- The four access control models
- Users, groups, and roles
- Privilege creep, and how to catch it

The access control models

Model	How access is decided	Seen in
DAC	The **owner** decides who may use it	Sharing a file or document
MAC	A **central policy** decides from labels	Military, high-security
RBAC	Access granted to **roles**; people get roles	Most businesses
ABAC	Decided by **attributes & rules** (dept, time, device)	Large cloud systems

You'll meet **DAC** (everyday sharing) and **RBAC** (the business backbone) most. Real systems blend them.

Users, groups, and roles

*Managing permissions per person doesn't scale. Organize access by **job, not person**.*

- ***User:** one account (a person, program, or device).*
- ***Group:** a bundle of users ("Phnom Penh Store Staff").*
- ***Role:** a named set of permissions for a job ("Cashier" takes payments, issues small refunds).*

*Decide once what a role may do; assign it to whoever does that job. Change jobs → change role → access follows. This is **RBAC**.*

Privilege creep

The slow build-up of permissions a person no longer needs.

- *Join sales → sales access. Help finance → finance access. Move to support → support access. Nobody removes the old ones.*
- *After years, one account can reach almost everything, a huge prize for an attacker.*

Fixes (process, not tech):

- ***Access reviews** (recertification): remove what's unused.*
- ***Joiner-mover-leaver:** grant, change, and promptly remove access.*
- ***Time-limited access** that expires automatically.*

Recap: models, roles, creep

- Four models: **DAC, MAC, RBAC, ABAC**; DAC and RBAC are the everyday ones.
- Manage access by **role**, not per person; changes happen in one place.
- **Privilege creep** rots least privilege over time; fix it with reviews and a leaver process.

Next lesson: all the kinds of security control, and how they stack into defense in depth.